

AC-DC Conversion

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003-1040559

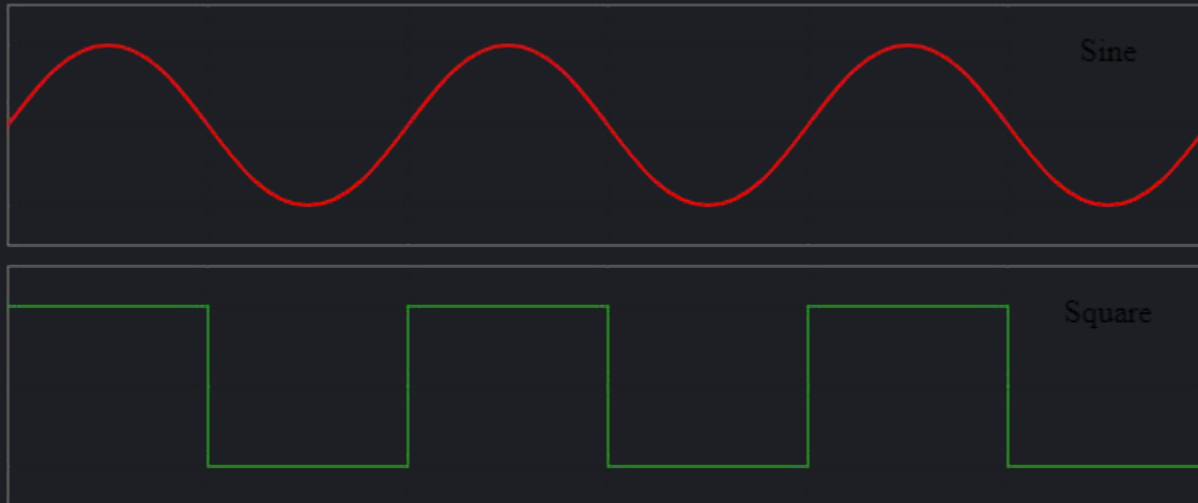
1250 003-77156.8

1760 0009-14563.7

73273

Objectives

1. Converts AC sine wave into a rectangular wave.
2. Output duty cycle depends on input amplitude.



Pulse Width Modulation (PWM)

1. Using duty cycle to control the intensity of an appliance

50% duty cycle

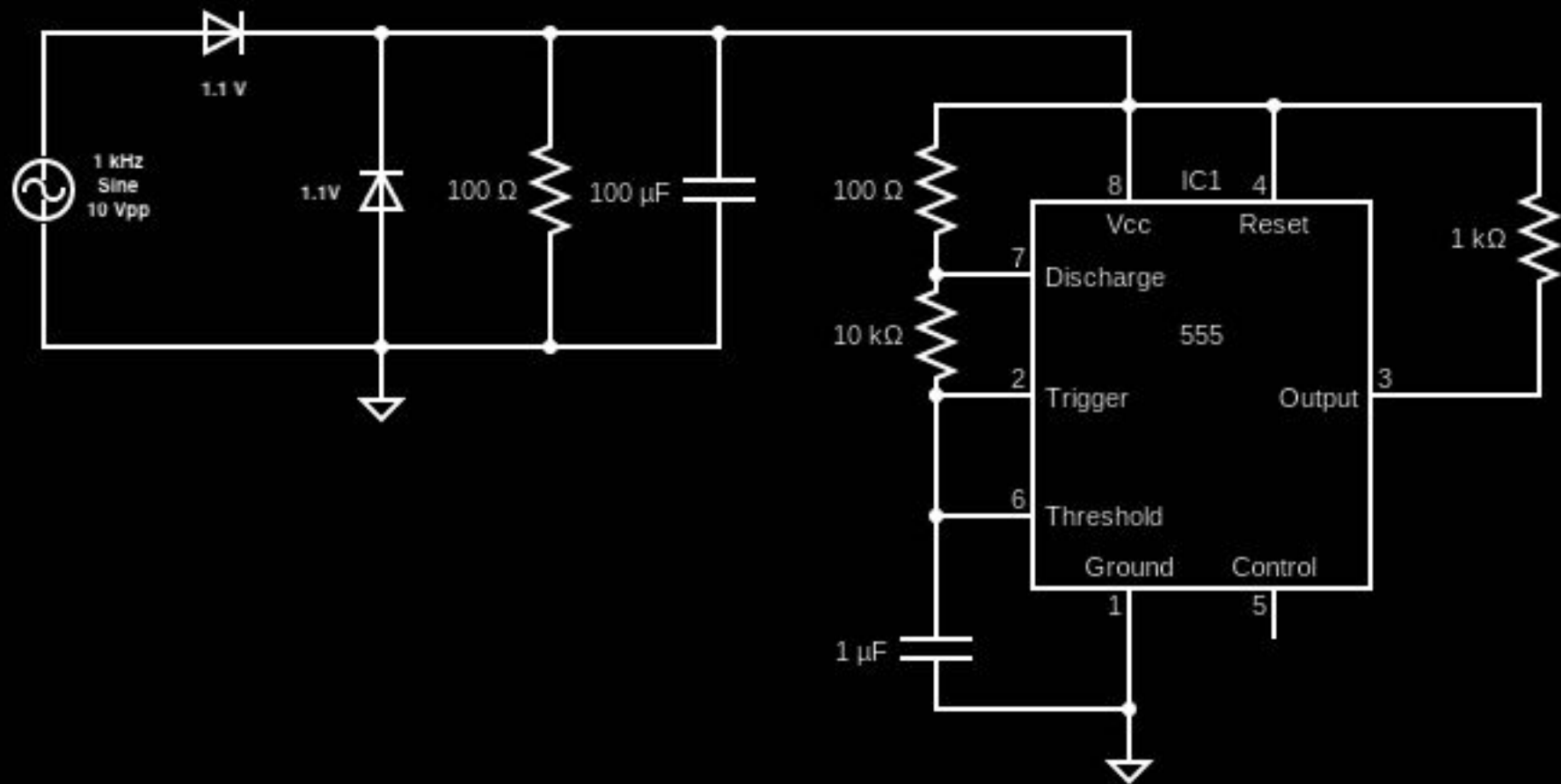


75% duty cycle



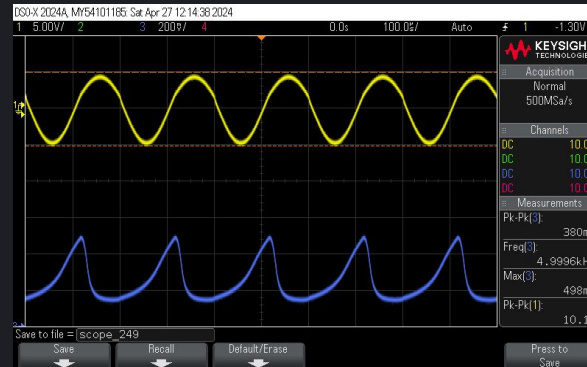
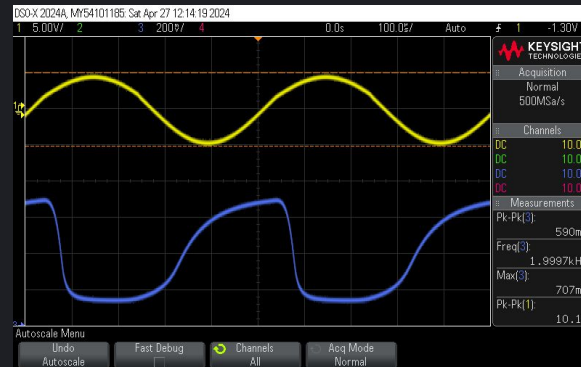
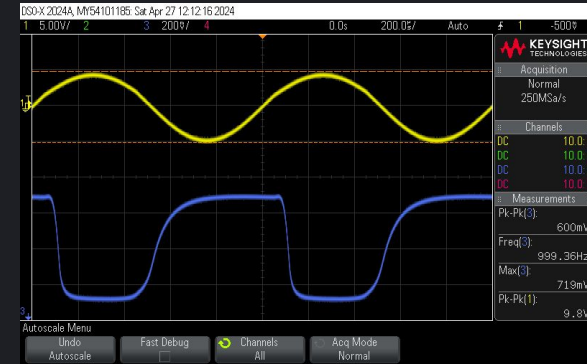
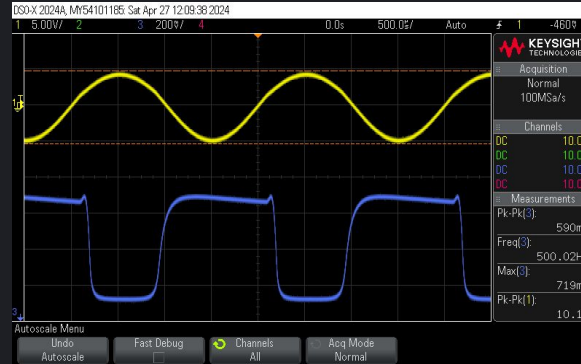
25% duty cycle





Comparing Frequencies

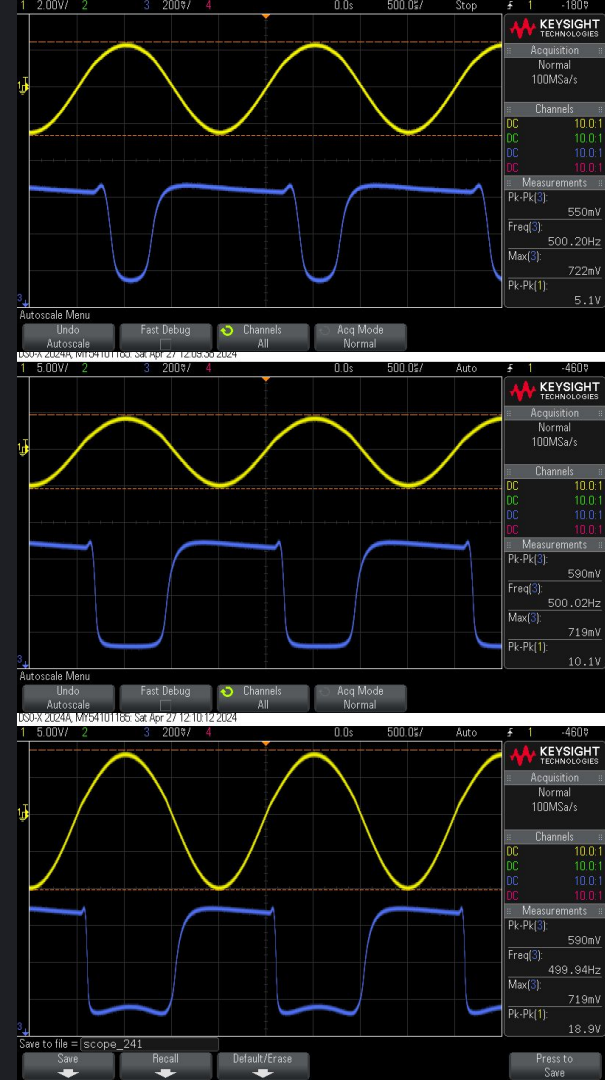
- 0.5, 1, 2, 5 kHz with 10 Vpp
- Works best at low frequencies
 - Sharper edges
 - Higher max voltage
- Low pass filter



500 Hz

- Max voltage constant at 720 mV
- Less square at higher duty cycles

Input Vpp	Duty Cycle
4	84.7 %
5	78.4 %
8	65.5 %
10	60.8 %
15	56.1 %
20	51.8 %

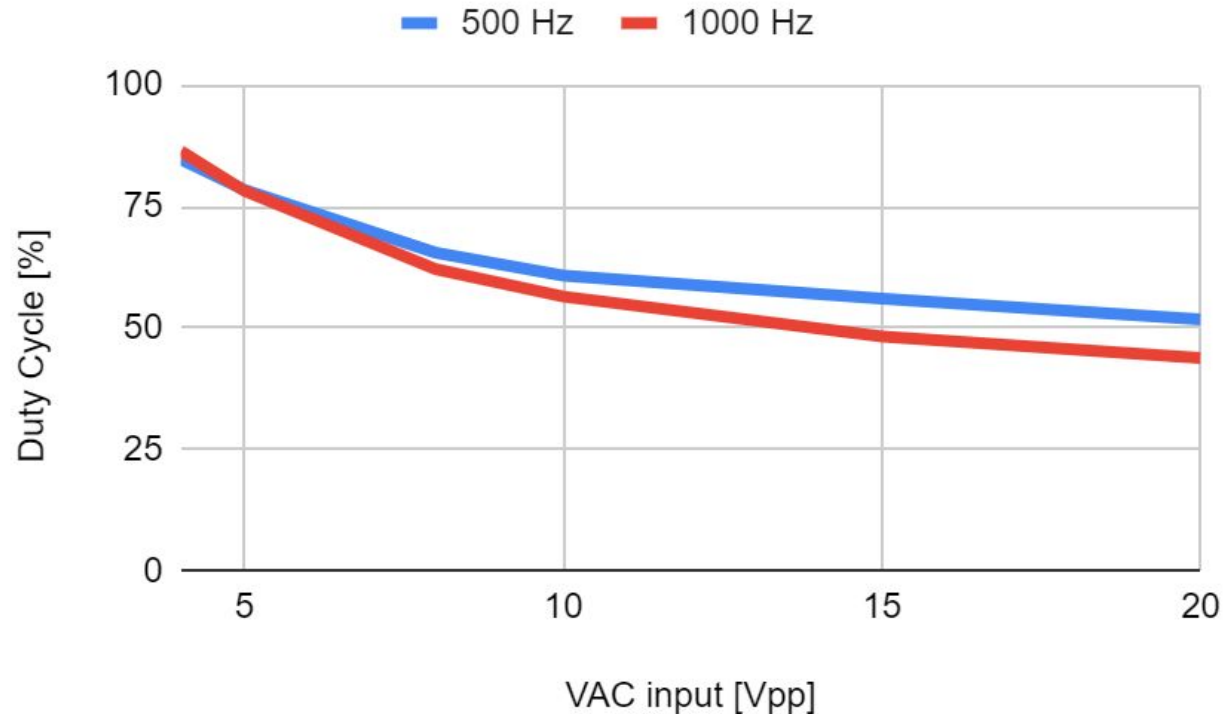


Duty Cycles

Objectives

1. Converts AC sine wave into a rectangular wave.
2. Output duty cycle depends on input amplitude.

V_{pp} - Duty cycle relationship can be changed with 555 timer circuit



Future Work

- Find frequencies with the flattest peaks.
- Higher output voltage for more applications.
 - AC-DC converter
 - Find resistor values for 555 timer
- Find resistor values with a wider range of duty cycles



01

02

03

04

05

06



Applications

1. Running a motor, LED's, heaters and more
2. Controlling power output without changing the maximum voltage

01

02

03

04

05

06





Thank you



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